

Multiple Choice (10 marks)

1. The maximum number of acute angles in a convex 10-gon in the Euclidean plane is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
2. Statement 1 | Every solvable group is of prime-power order. Statement 2 | Every group of prime-power order is solvable.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Find the degree for the given field extension $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$.
 - (a) 0
 - (b) 2
 - (c) 1
 - (d) 3
4. Find the maximum possible order for an element of S_n for $n = 7$.
 - (a) 6
 - (b) 12
 - (c) 30
 - (d) 105
5. Statement 1 | If a and b are elements of finite order in an Abelian group, then $|ab|$ is the lcm $(|a|, |b|)$. Statement 2 | If g is a group element and $g^n = e$, then $|g| = n$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The product of (2,3) and (3,5) in the ring $\mathbb{Z}_5 \times \mathbb{Z}_9$ is (3,1).
7. True or False: The circle defined by $x^2 + y^2 = 9$ intersects the parabola $x^2 = y + 4$ at the greatest number of points.
8. True or False: The area of an equilateral triangle whose inscribed circle has a radius of 2 is $12\sqrt{3}$ square units.
9. True or False: In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10, the identity element is 2.
10. True or False: The maximum possible order for an element of S_{10} is 30.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The inhabitants of the island of Jumble use the standard Kobish alphabet (20 letters, A through T). Each word in their language is 4 letters or less, and for some reason, they insist that all words contain the letter A at least once. How many words are possible?
12. Discuss the implications of Statement 1, which asserts that if H and K are subgroups of G and one is normal, then HK is a subgroup of G. Additionally, analyze Statement 2 regarding the abelian nature of groups of order p^2 , where p is prime. Evaluate the validity of both statements and their interrelation in group theory.

End of Paper